

EXPRIMENT-6

6.)Write a Python program for ciphertext has been generated with an affine cipher. The most frequent letter of the ciphertext is “B,” and the second most frequent letter of the ciphertext is “U.”Break this code.

PROGRAM:

```
1
2 import math
3
4 cipher1 = 'B'
5 cipher2 = 'U'
6
7 print("Possible Affine Cipher Keys:")
8
9 for a in range(1, 26):
10     if math.gcd(a, 26) == 1:
11         for b in range(26):
12             inv = pow(a, -1, 26)
13
14             p1 = ((ord(cipher1) - ord('A') - b) * inv) % 26
15             p2 = ((ord(cipher2) - ord('A') - b) * inv) % 26
16
17             if p1 == 4 and p2 == 19:
18                 print("a =", a, "b =", b)
19                 print("B maps to E and U maps to T")
20
```

OUTPUT:

```
Possible Affine Cipher Keys:
a = 3 b = 15
B maps to E and U maps to T
```